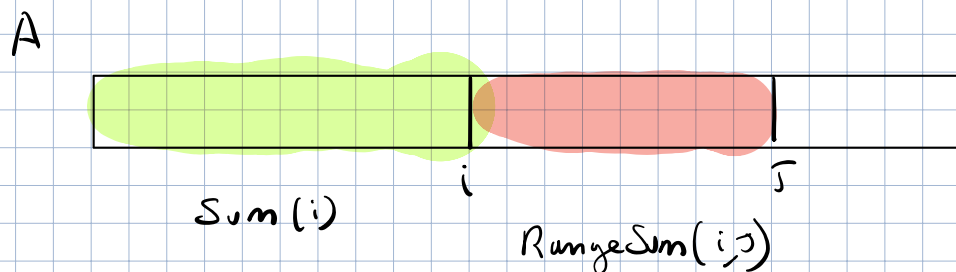


Static Prefix Search

Given $A[1, n]$ of integers, we would like to support:

- $\text{Sum}(i) = \sum_{k=1}^i A[k]$

- $\text{RangeSum}(i, j) = \sum_{k=i}^j A[k]$



$$\text{RangeSum}(i, j) = \text{Sum}(j) - \text{Sum}(i-1)$$

Solution

- Store P such that $P[i] = \sum_{k=1}^i A[k]$
- $\text{Sum}(i)$ takes constant time
- discard A

$$A[i] = \text{RangeSum}(i, i+1)$$

- P can be computed in $\Theta(n)$ time

Problem A1

Ilya has queries

Given a binary string $S[1, n]$ consisting in
a's or b's and a set of m queries

- $q(i, j)$ = returns the number of positions
 $k \in [i, j]$ such that
 $S[k] = S[k+1]$

Example

P	1	2	3	4	5	6	7
S	a	b	b	a	b	b	a

$q(2, 6) = 2$

Solution in $\Theta(n + m)$ time

P	1	2	3	4	5	6	7
S	a	b	b	a	b	b	a
B	0	1	0	0	1	0	0
P	0	1	1	1	2	2	2

$B[i] = \begin{cases} 0 & \text{if } S[i] \neq S[i+1] \\ 1 & \text{if } S[i] = S[i+1] \end{cases}$

$q(i, j)$ reduces to Range Sum (i, j)

Problem A2

Little girl and Queries

Given $A[1, n]$ and a set of m queries Q .

Each query is a RangeSum(i, j)

Our goal is to permute the values in A to maximize the sum of the results of the queries in Q .

Example

$$\begin{array}{cccccc} A & 5 & 2 & 4 & 1 & 3 \\ & \underbrace{\hspace{10em}} & & & & \\ & \underbrace{\hspace{2em}} & & & & \\ & \underbrace{\hspace{2em}} & & & & \end{array} \quad \begin{array}{r} 15 \\ + \\ 6 \\ + \\ 6 \\ = 27 \end{array}$$

$$\begin{array}{cccccc} A^\pi & 2 & 5 & 4 & 7 & 3 \\ & & & & & \\ & & & & & \end{array} \quad \begin{array}{r} 15 \\ + \\ 9 \\ + \\ 9 \\ = \boxed{33} \end{array}$$

Solution

- Step # 1

Compute the frequency of each entry

F	1	3	3	1	1
A	5	2	4	1	3

A'' 2 5 4 7 3

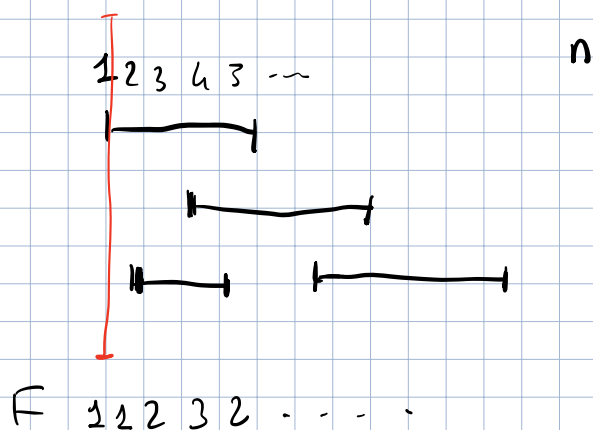
- Step #2

Sort by value and by frequency and match

$\Theta(n \log n)$ time

Focus Step #1

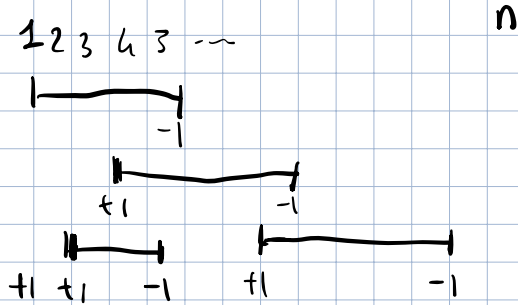
Solution #1



Sweep Line

- $\Theta(n + m \log m)$ time with comparison based sorting algorithm

- $\Theta(n + m)$ time with counting sort



C

	+1	0	+2	-1	
Sweep line	1	2	3	4	...
F	1	1	3	2	...

↑ Prefix Sums of C

$q(2, 5)$

$q(3, 6)$

$q(1, 2)$

C

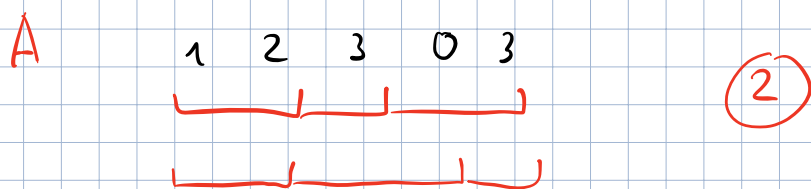
	1	2	3	4	5	6
		+1			-1	
			+1			-1
		+1	-1			
P	1	2	2	2	2	1

$\Theta(n + m)$ time

Problem #3

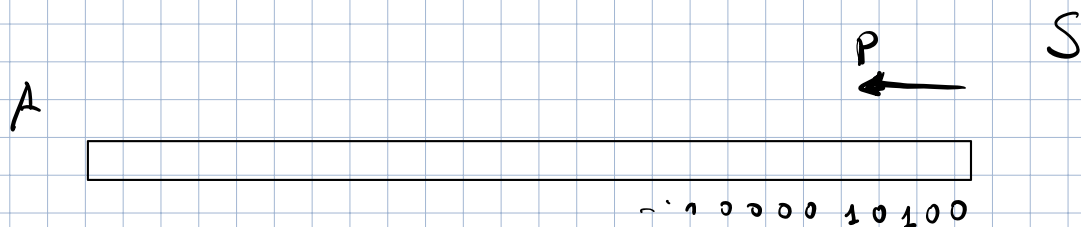
Number of Ways

Given $A[1, n]$, count in how many we can split A in 3 parts so that the sum of the parts is the same



Let's try with the easier problem of splitting in 2 parts

Solution for the easier problem

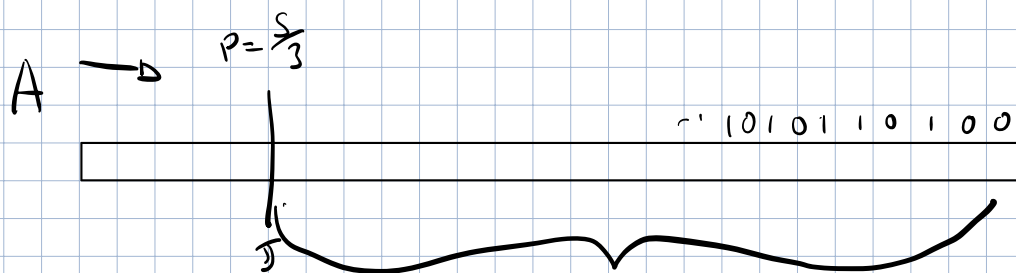


$\Theta(n)$ time

if $P == \frac{S}{2}$ then 1
else 0

Solution for original problem

- if $S \% 3 \neq 0$, no split



Sum is

$$2 \frac{S}{3}$$

how many ways to splitting of this?